

Lesson 2: Options from First Principles

Analyst onboarding · Lesson 2 of 10 · L/S Gamma Desk, QUANTT

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Goal: know exactly what an option is, what it pays, and why our desk trades them.

You need: Lesson 1 (SPY, long/short, bid/ask, volatility).

1 What an option is

An option is a contract that gives its buyer a **right, but not an obligation**, to buy or sell something at a price agreed today, up to a set date. The buyer pays the seller for that right up front. This payment is the **premium**.

Think of it like a concert-ticket reservation: you pay \$20 now to lock in the right to buy a \$100 ticket next month. If prices jump to \$300, your right is valuable. If they fall to \$50, you walk away and lose only the \$20.

- **Call option:** the right to **buy** at the agreed price.
- **Put option:** the right to **sell** at the agreed price.
- **Strike price (K):** the agreed price.
- **Expiry:** the last day the right exists. **DTE** means *days to expiry*.
- **Exercise:** using the right, i.e. actually buying or selling at the strike.
- **Underlying:** what the option is on. For us, **SPY**.

⇒ The buyer risks only the premium and keeps the upside. The seller collects the premium and takes on the risk.

2 The details of SPY options

- **One contract covers 100 shares.** A quoted price of \$13.40 costs $13.40 \times 100 = \$1,340$.
- **American style:** they can be exercised on any trading day up to expiry.
- **They trade with a bid and an ask** like SPY itself, and the spread is a cost.
- **There are expiries for every trading day.** Our desk uses Friday expiries about one month out.

3 What an option pays at expiry

On the expiry day the option is worth only what exercising it would pay:

$$\text{Call value} = \max(S - K, 0) \quad \text{Put value} = \max(K - S, 0)$$

Here S is SPY's price at expiry and "max" means "take the larger of the two". If exercising would lose money, you simply don't, so the value is 0. **Profit = value – premium paid.**

Example. SPY trades at 760. The 761-strike call expiring in 30 days costs \$13.40, and the 761 put costs \$11.20 (real fills from our paper account).

SPY at expiry	Bought the call	Bought the put	Sold the put
740	−\$1,340	+\$980	−\$980
761	−\$1,340	−\$1,120	+\$1,120
780	+\$560	−\$1,120	+\$1,120

- **Bought the call:** the most it can lose is \$1,340. It breaks even if SPY ends at $761 + 13.40 = 774.40$.
- **Bought the put:** the most it can lose is \$1,120. It breaks even at $761 - 11.20 = 749.80$.
- **Sold the put:** it keeps the whole \$1,120 if SPY ends above 761, and loses more the further SPY falls.

⇒ Buying options is a bet on a **big** move. Selling options is a bet on a **small** move. This one idea drives everything our desk does.

4 Where the strike sits: moneyness

- **At the money (ATM):** the strike is near SPY's current price. Our desk trades ATM options.
- **In the money (ITM):** exercising now would pay something (a call with K below SPY, a put with K above).
- **Out of the money (OTM):** exercising now would pay nothing (a call with K above SPY, a put with K below).

5 Why an option is worth more than its payoff today

Before expiry, an option's price has two parts:

- **Intrinsic value:** what it would pay if exercised right now. The 761 call with SPY at 760 has none.
- **Time value:** everything else. The call still costs \$13.40 because SPY might rise a lot in 30 days.

Time value melts to zero by expiry. How much the market charges for it depends on **how much it expects SPY to move**, i.e. on volatility. Lesson 3 turns that into a number.

6 Buyers and sellers face very different risks

	Buyer (long the option)	Seller (short the option)
Premium	Pays it	Receives it
Worst case	Loses the premium	Large losses (unlimited when selling calls)
Hopes for	Big moves	Small moves
Each day that passes	Costs money	Earns money

⇒ Sellers earn small, steady income and sometimes take a big loss. Buyers pay a small, steady cost and sometimes win big. Choosing **when to be which** is our job.

7 Why use options at all?

1. **Curved payoffs:** gains grow faster as the move grows, while losses are capped at the premium. Shares pay in a straight line.
2. **A bet on movement itself:** options let you bet on *how much* SPY moves, not *which way*. Lesson 5 shows how we remove the direction completely.
3. **Leverage:** \$1,340 gives exposure to 100 shares worth about \$76,000. That cuts both ways.

Our desk uses options for reason 2: we are **volatility traders**, not directional traders.

8 Words to know

- **Call / put:** right to buy / sell. **Strike:** the agreed price. **Premium:** the option's price.
- **Expiry, DTE:** last day, days left. **Exercise:** use the right.
- **ATM / ITM / OTM:** strike near / in favour of / against the current price.
- **Intrinsic / time value:** payoff now / the extra the market charges for possible movement.

9 Check yourself

1. You buy the 761 put for \$11.20 and SPY finishes at 755. What do you make or lose on one contract?
2. Why is the 761 call worth \$13.40 when exercising it today pays nothing?
3. If SPY doesn't move at all for 30 days, who makes money: the call buyer or the put seller?

Answers: (1) $(761 - 755 - 11.20) \times 100 = -\520 . (2) Time value: SPY might rise a lot before expiry. (3) The put seller keeps the premium; the call buyer loses it.